Explicit Euler

First-order forward discretization

1 Numerical map

For an initial-value problem

$$\dot{z} = f(t, z), \qquad z(t_n) = z_n,$$

Taylor expansion gives

$$z(t_n + h) = z_n + hf(t_n, z_n) + \mathcal{O}(h^2).$$

Explicit Euler discards the remainder and defines

$$\Phi_{h,t_n}(z_n) = z_n + hf(t_n, z_n).$$

The method requires one vector-field evaluation, has local truncation error $\mathcal{O}(h^2)$ and, under the standard stability and smoothness assumptions, global error $\mathcal{O}(h)$.

2 Stability

For the test equation $y' = \lambda y$ the amplification factor is

$$R(\xi) = 1 + \xi, \qquad \xi = h\lambda.$$

Absolute stability requires $|1 + \xi| \leq 1$, the disk centered at -1 with radius one. The bounded stability region makes this method unsuitable for stiff problems unless the step is very small.

3 Hamiltonian interpretation

For a canonical Hamiltonian system with Jacobian $A = Df(z_n)$, the step tangent is $D\Phi = I + hA$. Even when A is Hamiltonian, $(D\Phi)^T \Omega D\Phi = \Omega + h^2 A^T \Omega A$ in general. Explicit Euler is therefore neither symmetric nor symplectic and does not preserve energy in general. It serves as a transparent first-order baseline for accuracy and geometric diagnostics.

4 Implemented configuration

`ExplicitEuler` has no numerical configuration variants beyond shared progress and step-observer controls. It consumes the general `DynamicalSystem` contract, performs no nonlinear solve, and publishes only the number of accepted main steps. Requested off-grid output values are computed by disposable shadow steps from the preceding main node, so output density does not change the fixed-grid trajectory.

5 Implementation correspondence

The method is implemented in `src/simulation/methods/classical/euler.py`. The fixed-grid execution lifecycle is documented in the sibling `simulation/` directory; shared physical dynamics are documented outside the model tree.